SECURITY ENGINEERING RESEARCH POSTER

Reproducible Regression Harness for Machine-Learning Security Tools

Contract-Validated Adapters Catching Cross-Repo Breakage Before It Ships

github.com/poojakira/mlsec-benchmark-suite (public, default branch main, primary language Python). Apache-2.0 at this historical commit (later revisions may use other terms) • Python 3.12 • HEAD 5c20f1da9602eb162efe8c63158da52b11d61f60 • verified 2026-10-01

EVIDENCE POSTURE

Verified, scoped, reproducible

Claims are constrained to the repository evidence snapshot and stated limitations.

1.00	3/0	83 across 10 modules	82 passed, 1 skipped
HF-SCANNER F1	TP / FP	Test functions	Historical pinned CI

SYSTEM VIEW

Architecture and Threat-Control Flow

Trust Boundary

Input surface
Data, models, requests, policies, or artifacts entering the tested path.

Control surface
Repository-specific validation, analysis, and fail-closed checks.

Evidence surface
Findings, metrics, tests, logs, or reproducible artifacts.

This diagram describes the tested repository path, not a universal deployment guarantee.

01 **Inspect the trust boundary** →

02 **Apply repository-specific security checks** ↓

03 **Record reproducible evidence**

Research Contribution

A contract-validated regression harness that wraps ML security tools in typed in-process adapters, runs them against versioned fixtures, and fails pytest on schema drift - with a hard, labeled separation between REAL product adapters and a synthetic smoke-test scaffold.

Verified Evidence

EV-01	Statement coverage: 90.07%	**EV-02**	Real fixtures: 5
EV-03	Real adapters: gated CI	**EV-04**	Known-bad flagged: 100
EV-05	Known-good clean: 100	**EV-06**	Smoke scaffold (synthetic): 50
EV-07	VERIFIED_AT_SNAPSHOT - Real HF-scanner adapter: precision=recall=F1=1.0 on 5 fixtures - README verified real run; tests/test_hf_scanner_adapter.py end-to-end (no mock); total_tp=3,total_fp=0,total_fn=0.	**EV-08**	VERIFIED_AT_SNAPSHOT - 83 test functions across 10 test modules; current Python 3.12 CI recorded 82 passed, 1 skipped with 90.07% statement coverage - repository README and test tree at HEAD 5c20f1da9602eb162efe8c63158da52b11d61f60.
EV-09	VERIFIED_AT_SNAPSHOT - JSON-schema contracts + versioned fixtures; pytest fails on drift - README how-it-works; schemas/ + contracts/ present.	**EV-10**	UNSUPPORTED (disclaimed by repo) - run-smoke per-category numbers - README: seeded-PRNG synthetic scaffold for plumbing, NOT measurements. Shown as synthetic only.

Method and Control Path

01 **Inspect the trust boundary**

02 **Apply repository-specific security checks**

03 **Record reproducible evidence**

Limitations and Residual Risk

- Validates contracts/format, not real-world accuracy.
- run-smoke numbers are synthetic (seeded PRNG).
- Fixture coverage is finite; gaps possible.
- Adapters can go stale if a tool changes internals.
- Real adapter gating requires sibling install.

Reproducibility

```
Evidence: results/real_hf_scanner.json, METHODOLOGY.md,
tests/
```

Evidence Provenance

Claim discipline
Metrics come from the repository's verified metrics and research brief.

Scope discipline
Benchmarks are reported only at their documented fixture, dataset, or test scope.

Reproduction
PDF and PNG are generated locally from committed evidence without paid services.

Evidence-backed security engineering

README, verified metrics, tests, CI, and committed artifacts remain authoritative. Poster layout emphasizes architecture, controls, evidence, limitations, and reproduction rather than unsupported efficacy claims.

36 x 48 in
PDF + PNG parity